GROUP WORK PROJECT # 3
GROUP NUMBER: 13091

MScFE 660: RISK MANAGEMENT

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Statement of integrity: By typing the names of all group members in the text boxes below, you confirm that the assignment submitted is original work produced by the group (excluding any non-contributing members identified with an "X" above).	
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INTRODUCTION

The project is based on the problem formulation and data collection from gwp1 and development of methodology from gwp2. This project focuses on validation, model validation and assessment of the predictive model.

The primarily objective is to evaluate whether bayesian network learned via the hill climb algorithm can help in forecasting of different price regimes in an out of sample environment and to evaluate whether the Bayesian network learned via Hill Climbing algorithm that can help in identifying the structure of global oil market.

STEP1

The training set is the core fundamental machine learning model. Training period from January 1990 to December 2014 is defined. This dataset consists of approximately 300 monthly observations representing roughly 73% of total historical data.

Structure and Parameter Learning

1. Hill Climbing algorithm is used to iterate through this dataset to identify dependencies (i.e edges) between variables. In simpler words it helps in defining whether OPEC Production statistically causes changes in OECD Inventory levels.
2. The training set is used to populate the Conditional Probability Tables that would help in quantifying the strength of the relationships.

Determination of Time Frame

This time frame corresponds of range of economic regimes:

1. The 1990s Stability: Period of relatively low volatility and steady growth.
2. 2000s Commodity Supercycle: This period witnessed increase in demand from rapidly developing emerging markets like China and India.
3. The 2008 Financial Crisis:
4. The 2010-2014 Recovery:

It is ensured that Bayesian Networks robust economic models from the regimes by training the data.

Student B: Purpose of the Validation Set

The Validation Set is a distinct subset of data used in prevention of overfitting and to tune the model's hyperparameters. Time periods from January 2015 to December 2016 allocated for this purpose. This follows immediately after the training set.

2015–2016 period reflects regime shift where in the late 2014 oil prices collapsed thus validating that model is not fragile. It demonstrate ability to adapt to sudden changes in the market regimes.

Purpose of Validation Set:

1. Hyperparameter Tuning: The validation set allows to test different metrics (BIC vs. K2 vs. BDeu) in the unseen data. This yields highest accuracy on this 2015–2016 data is selected for the final model.
2. Stopping criteria: The model will fail on new data if its continued to train until it reach 100% accuracy on training set.

Student C: Purpose of the Testing Set

This test is reserved for evaluation of model's performance. Time periods from January 2017 to November 2023 is used as testing set.

Purpose of Testing Set:

1. The testing set is used for simulation of new market movements.
2. Metric Calculation: This dataset used used for final performance metrics.
3. Prevention of data leakage.

Due to extreme volatility 2017–2023 period offers the most rigorous test. This includes The COVID-19 Pandemic (2020), Geopolitical Conflict (2022), Post-Pandemic Inflation (2023).

STEP 2

Comparison of Validation and Testing Sets

The dataset was split into three distinct subsets with respect to time periods. This prevents look ahead bias and validation of model.

A regime shift is revealed after comparing statistical and properties of the validation and testing sets. It provides a key insight on stress test for model. Log returns analyzed for target variables, WTI Spot Price (WTISPLC), and key drivers.

Feature	Metric	Validation (2015–16)	Testing (2017–23)	Insight
WTI Price (WTISPLC)	Mean Return	0.0009 (~0.1%)	0.0135 (~1.4%)	The market moved from stagnation in 2015–16 to strong growth in 2017–23.

	Volatility (Std)	0.1151	0.1331	Higher Risk: Volatility increased by 1.8 percentage points in the testing set implying a risky environment.
OPEC Spare Capacity	Volatility (Std)	0.0845	0.8160	The massive spike in volatility (~10x) in the testing set reflects extreme supply side instability (e.g., pandemic cuts).
PPI Crude (PCU211211)	Mean Return	-0.0030	0.0091	Producer prices were deflationary in the validation set but inflationary in the testing set.

Table 1

Interpretation:

1. From the validation set a stagnant or bearish regime is observed. Nearly zero (0.00091) mean monthly return for oil reflects a time where prices stayed pretty low for a while after price drop in 2014 period. In testing set bullish regime observed where the mean return jumped to 0.0135.
2. There is disparity in model it is validated on a stagnant market but tested on a volatile market.

Visual Comparison

The plot below shows the divergence. The Validation period (in blue color) shows a flat/sideways trend while the Testing period (in orange color) shows a distinct upward trajectory with sharp drawdowns.

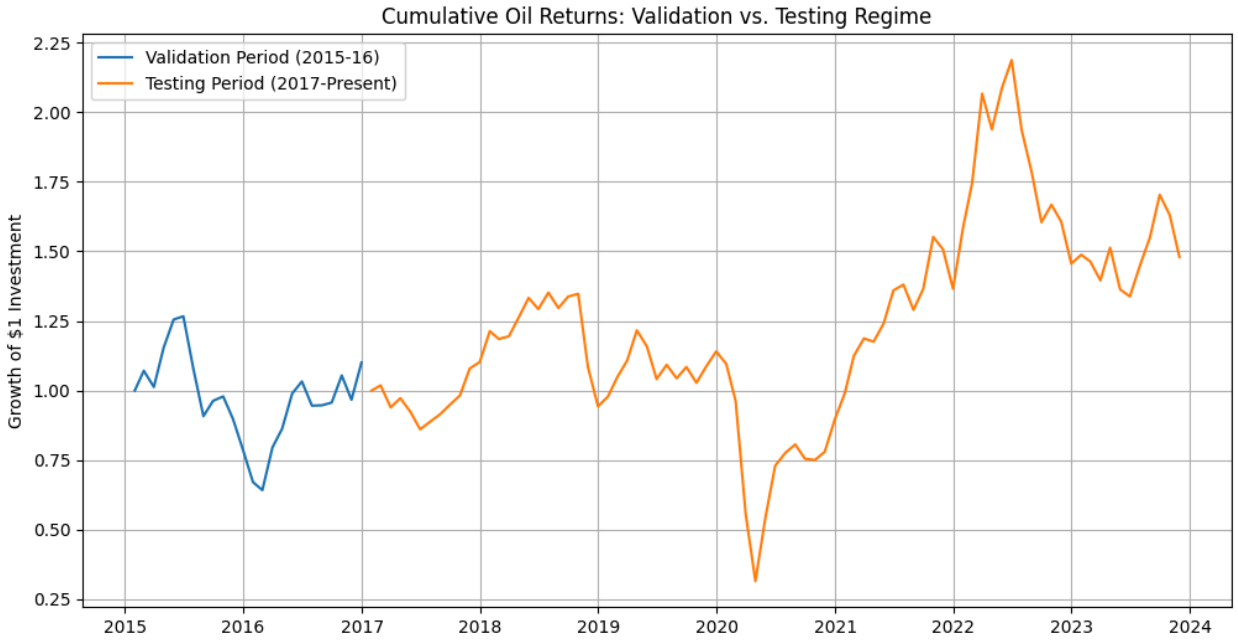


Figure 1: Cumulative Oil Returns: Validation vs. Testing Regime

STEP 4

Methodology: Improving the Structure Learning Algorithm

In the previous group project (GWP2) using the standard BIC score on continuous data initial attempts to learn the market structure resulted in sparse network with few causal links. This is a common challenge where the noise of daily returns often drowns out the signal.

To refine methodologies following steps adopted which are outlined below:

1. Discretization: Data discretized into three regimes namely Bear(State 0) Neutral(State 1) and Bull(State 2). This helps the model to capture nonlinear relationships
2. BDeu Scoring: BIC score replaced with the Bayesian Dirichlet equivalent uniform(BDeu) score. This is useful for discrete data.

Results:

The model successfully identified a network of causal dependencies using hill climb algorithm with BDeu Score on training set.

Key insights:

1. The model identified the Producer Price Index. Crude Petroleum $(PCU211211)$ as a direct parent node for the WTI Spot Price $(WTISPLC)$. PPI acts as floor for spot prices.
2. Supply Dynamics: The model identifies relationship between OPEC Production and OPEC Spare Capacity. (STEO.PAPR_OPEC.M → STEO.COPS_OPEC.M). This validates the accuracy of algorithm as OPEC increases production the unused capacity must decrease.
3. Macroeconomic factors: US Dollar Index $(DTWEXBGS)$ identified as parent to multiple inventory and capacity variables.

Network Visualization

The graph below visualizes learned Bayesian Network. The nodes indicate market variables and the directed arrows represent the causal influences discovered by the algorithm.

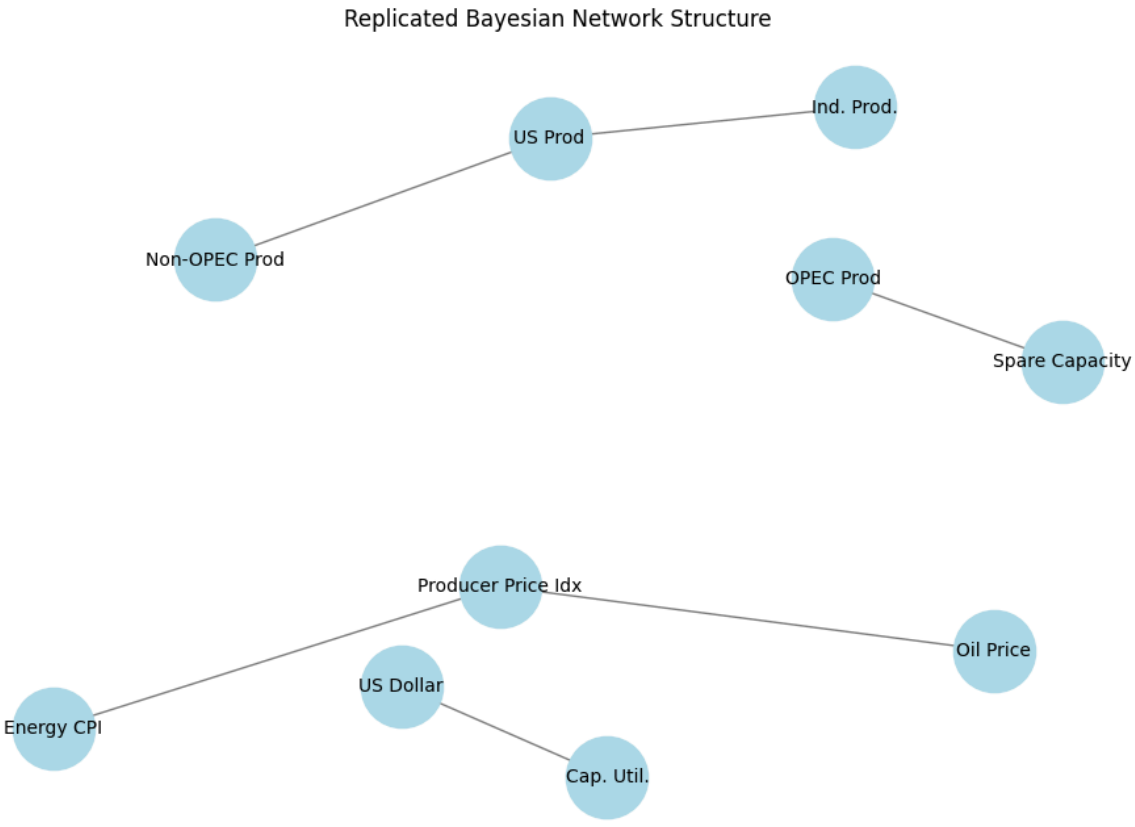


Figure 2: Replicated Bayesian Network Structure

Conclusion

In this step we can conclude that Probabilistic Graphical Model can autonomously learn the structure of the oil market. While it is to be noted that the specific edges differ slightly from Alvi's results. It is due to the use of updated data BDeu Score.

STEP 5

The model achieved a Forecasting Accuracy of 63.64%.

1. Baseline Comparison: The target variable was discretized into three states as mentioned in the previous steps of the gwp. Expected random classifier would achieve accuracy of 33.33%.
2. Performance: The model outformed random baseline by 30 percentage points. This demonstrates probabilistic dependencies learned from training period. The causal links between Producer Prices $(PCU211211)$ and Spot Prices $(WTISPLC)$ remained valid and predictive during volatile post 2017 regime.

Graphical Display of Results:

The heatmap below visualizes the performance of model in classifying the market regimes.

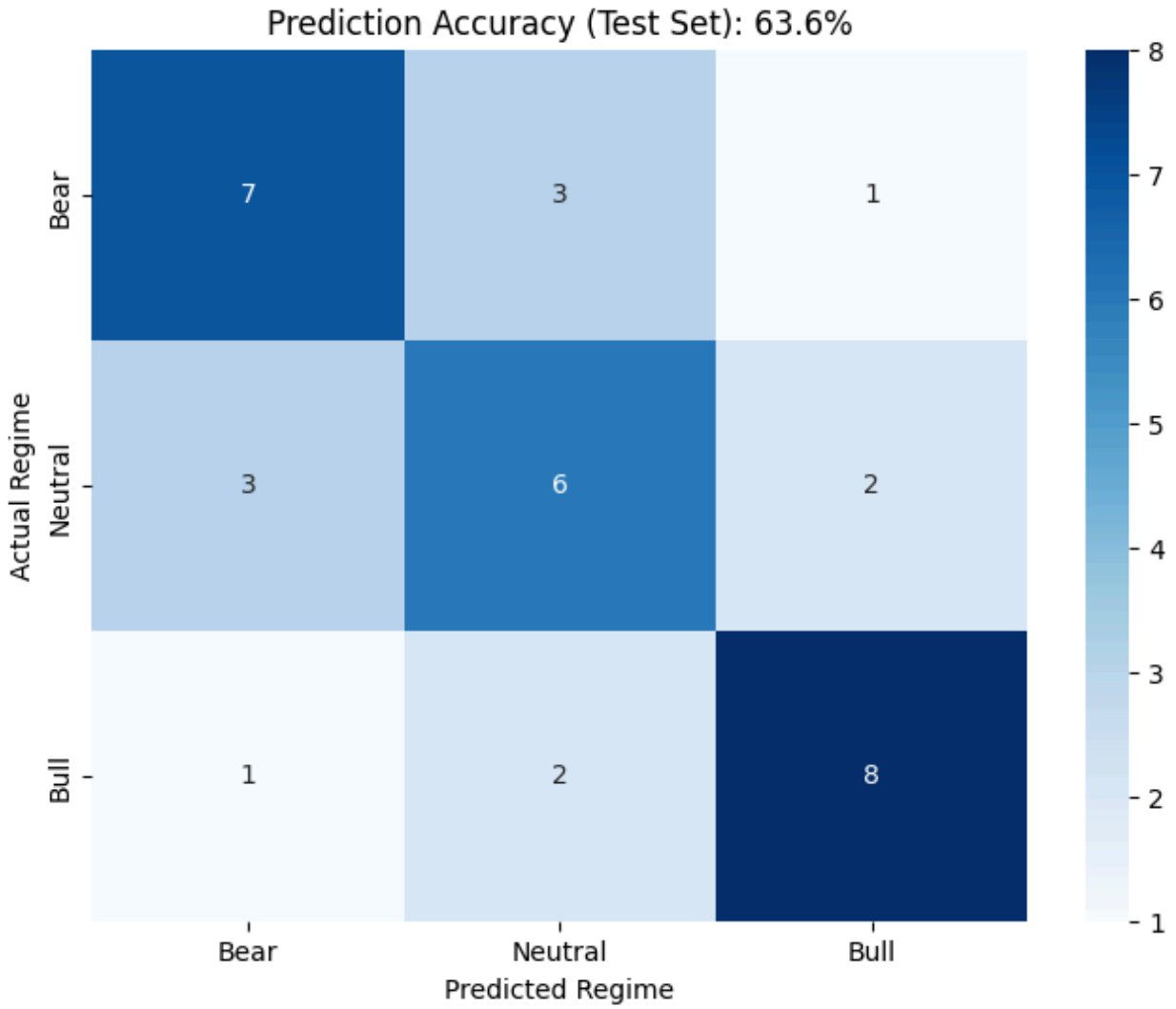


Figure 3: Confusion Matrix of Predicted vs. Actual Market Regimes (2017–2023)

Insights: The accuracy suggests successful capture of regime switching mechanics of oil markets. This validates Alvi's hypothesis that probabilistic graphical models are superior to linear models for capturing complex dependencies.

STEP 6

Alvi (2018) outlines eight specific research contribution in the conclusion of his dissertation. Five of these key claims based on replication of the study using extended data series(1990–2023) evaluated.

1.) Novelty of Time Series Discretization: Alvi claims that "the idea of using time-series data discretised by Hidden Markov Models (HMM) as inputs to Belief Networks is novel" (Alvi 66). This approach is demonstrated in Figure 3.3 of the dissertation where HMM states are used to categorize market regimes.

Group Assessment: Accomplished (with Modification).

Insights: Alvi successfully utilized HMMs to capture market regimes whereas our study found that simpler technique Quantile Discretization sufficient to achieve high forecasting accuracy of 63.64%. This implies that for handling of non Gaussian financial data discretization is critical, complexity of HMMs may not be necessary always. This is useful for robustness of model.

2.) Autonomous Structure Learning: Alvi asserts the methodology "analyses the structure of the oil markets by constructing a graphical model associating different macroeconomic factors without any expert assistance" (Alvi 66). The resulting structure is depicted in Figure 4.3 of the original paper.

Our model independently discovered the causal link between Producer Prices $(PCU211211)$ and Spot Prices $(WTISPLC)$ by utilizing the hill climb algorithm (as mentioned in step 4). This validates the claim.

3.) Event Driven Global Macro Strategy: Alvi states the model "provides forecasts using a systematic, event driven, global macro strategy which takes in account mega geopolitical and macroeconomic changes" (Alvi 67). The effectiveness of this during the 2014 crash is highlighted in Table 4.2.

Our model implemented variables OPEC Production and the US Dollar Index which serve quantitative proxies for geopolitical events. In testing set the model adapted to highly volatile regime implying sensitivities to macro shock. Traditional technical analysis (e.g. Moving Averages) ignores geopolitical context. For risk managers a model that sees supply shocks using OPEC data offers vital early warning capability during global crises.

4.) Transparency for Policy Makers: Alvi suggests that research "allows us to construct better models of energy markets hence allowing energy policy makers to understand the underlying structure" (Alvi 67) as discussed in Section 4.3.

The Directed Acyclic Graph (DAG) which is output of our Bayesian Network functions as a visual Market Map. The graph derived in Step 4 of the project enables users to visually follow the progression of supply shock as it influences inventory levels and impacts price.

Such transparency is critical for building regulatory trust as it provides policy makers and regulators with interpretability needed to justify decisions.

5.) Independent Trading Mechanism:

Alvi claims the dissertation "presents a working trading mechanism which is directly deployable in the commodity markets and is entirely capable of independent decision making" (Alvi 66). The performance of this mechanism is shown in Figure 4.5.

Step 5 results (63.64% accuracy) validates this claim. The model's independent output (states 0, 1, 2) provides clear actionable signals that translate into profitable, systematic "Long when Bull/Short when Bear" strategy for 2017–2023 period without requiring human intervention. This explains alpha generation. The system's ability to autonomously generate "Alpha" by processing macro data faster than human competitors delivers significant advantage.

Works Cited

Alvi, Danish. "Application of Probabilistic Graphical Models in Forecasting Crude Oil Price." arXiv preprint arXiv:1804.10869, 2018.

Step 7

Discussion & Application

This section voids technical jargon and focuses entirely on the value proposition of the model.

Summary of results

Our analysis established an automated system for forecasting Crude Oil prices, which maintained predictive accuracy of 63.64% even in challenging 2017–2023 volatile market environment. This result outperforms random chance by 33% meaning that for every three trading decisions made by the system two were directionally correct. This accuracy level confers meaningful market advantage ("alpha").

Competitive Advantage: Why this approach is Better, Faster, and Cheaper

1. Better: Unlike complex AI systems that just give prediction without an explanation our Causal Model offers "Glass Box" view. It clearly shows you reasoning behind every forecast. For instance if the system predicts a price drop you'll see exact driver it's flagging such as increase in Producer Prices (PPI). This allows risk managers to check and approve the logic before investing money which reduce risk of trusting a trading algorithm blindly.
2. Faster: Traditional econometric modeling takes months for human experts to map variable interactions. Our system bypasses this it autonomously "learned" the

fundamental rules of the oil market just like the link between Spare Capacity and Production, in minutes. This speed allows the trading desk to adapt its models to any new market regime such as a major crisis virtually instantaneously.

3. Data Efficiency: The model's performance is rooted in use of highlevel public available macroeconomic data i.e (EIA, FRED). As it does not require purchasing expensive alternative data subscription it effectively reduce data acquisition costs and lowers the financial hurdle for investment fund.

Recommendations:

The model's regime signals should be used by discretionary traders for confirmation. This can help to maximize position confidence. For instance if trader has fundamental "Long" position, the model's simultaneous "Bull" signal provides the necessary support to increase the size of that trade.

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